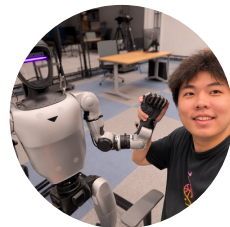


Chongyang Xu (许重阳)

🏠 Homepage 📍 College of Computer Science, Sichuan University (Wangjiang Campus)

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I'm a Ph.D. student passionate about **Physical AI and Robotics**. My research journey evolved from human motion capture and social behavior recognition into embodied AI, now focusing on **world action models, humanoid control, and loco-manipulation**.



Available for 6–18 month full-time internships worldwide. I'm passionate about hands-on debugging robots (Unitree G1, ALOHA, etc.), developing algorithms, and end-to-end deploying systems. Eager to contribute to realizing Physical AGI.

EDUCATION

Sichuan University, Chengdu

Ph.D. student in Computer Science and Technology

B.Eng. in Software Engineering

Advisor: Prof. Ziliang Feng

Sept. 2023 – Present (Expected Graduation: Jun. 2028)

Sept. 2019 – Jun. 2023

INTERNSHIP EXPERIENCE

Tongyi Robotics, Alibaba Group (通义具身实验室) Embodied Intern, working with Yilun Chen Feb. 2026 – Present

Research on **world action models (WAM)** and **humanoid loco-manipulation**; WAM foundation model pretraining on **128×H100**. Hands-on debugging **Unitree G1**, reproducing SONIC-based VLA for loco-manipulation. Independently won **1st Place (¥1M prize)** in Physical AI Competition (Embodied Track), G1 Humanoid Loco-Manipulation Door-Opening.

Dexmal (原力灵机) Research Intern, mentored by Shuaicheng Liu

Mar. 2025 – Feb. 2026

Reproduced VLA policies (π -series) and researched bimanual manipulation with 3D representations (**ICRA 2026, CVPR 2026**); won **1st Prize** in RoboTwin Challenge (CVPR 2025). End-to-end data collection, algorithm design, and real-robot deployment on ALOHA. Co-designed RoboChallenge; tuned X-VLA on 48 H20 GPUs, **0.9B surpassing π_0 (3B)**.

MEGVII (旷视科技) Research Intern, mentored by Shuaicheng Liu

Sept. 2024 – Mar. 2025

Built simulation pipeline integrating PyBullet with Blender for deformable objects (fluids, soft bodies, cloth). Developed multimodal IL + RL pipeline for contact-rich manipulation; won **1st Place** in ManiSkill-ViTac Challenge (CVPR 2025).

COMPETITIONS

1st Place - Physical AI Competition, Embodied AI Track

Shenzhen AI Industry Association, Apr. 2026

Unitree G1 humanoid autonomous door-opening based on SONIC locomotion control; awarded ¥1M shared prize.

1st Place - ManiSkill-ViTac Challenge Track 1 & 3

Embodied AI Workshop, CVPR 2025

Multimodal visuotactile IL + RL for closed-loop control; optimized tactile sensor geometry.

1st Prize - RoboTwin Challenge (Sim Round 1 & Real-World Track)

MEIS Workshop, CVPR 2025

Dual-arm VLA models (π_0 , RDT) for bimanual collaboration in simulation and real-world deployment.

PUBLICATIONS

Action-Geometry Prediction with 3D Geometric Prior for Bimanual Manipulation.

Full list at [Homepage](#)

Chongyang Xu, Haipeng Li, Cheng Shen, Haoqiang Fan, Ziliang Feng, Shuaicheng Liu.

Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR), 2026.

HeRO: Hierarchical 3D Semantic Representation for Pose-aware Object Manipulation.

Chongyang Xu, Cheng Shen, Haipeng Li, Haoqiang Fan, Ziliang Feng, Shuaicheng Liu.

IEEE International Conference on Robotics and Automation (ICRA), 2026.

Multi-Scale Hypergraph Relational Reasoning for Weakly Supervised Recognition of Group Activities.

Chongyang Xu, Runtian Zheng, Ziliang Feng, Chengfang Zhang.

IEEE International Conference on Multimedia and Expo (ICME), 2025. (Oral presentation)

RoboChallenge: Large-scale Real-robot Evaluation of Embodied Policies.

Adina Yakefu, Bin Xie, Chongyang Xu, Enwen Zhang, Erjin Zhou, et al.

arXiv preprint arXiv:2510.17950, 2025. **First Large-scale Real-world Robot Benchmark**

SKILLS

- **Robotics:** Hands-on debugging, calibration, and real-robot deployment across diverse platforms: **Unitree G1** (humanoid locomotion and loco-manipulation), **ALOHA** (Cobot Magic, DOS-W1 bimanual), **Franka**, **UR5**, **ARX5**; proficient in teleoperation data collection, joint-space/Cartesian pose control, ROS/DDS communication, and sensor integration.
- **Algorithms:** VLA foundation models (π -series, RDT, X-VLA), diffusion policies (DP, DP3), ACT, imitation learning, reinforcement learning, flow matching; 3D foundation models (VG GT, π^3), DINOv2/Stable Diffusion feature lifting into 3D semantic fields; SONIC-based humanoid locomotion control; human mesh recovery and multi-person 3D reconstruction.
- **Engineering:** End-to-end pipeline from data collection and curation, algorithm design, distributed training, to real-robot deployment and evaluation; PyTorch distributed training on large GPU clusters (**128×H100** for foundation model pre-training, 48×H20 for fine-tuning); large-scale hyperparameter search and ablations; PyBullet + Blender physics simulation, MuJoCo and Isaac Sim for locomotion policy training; server-client multi-robot deployment; Linux, Docker, Git, Wandb.